

CRYPTOGRAPHY

Security, Attacks, Goals and Evolution of Substitution Ciphers

CSA5112 – Cryptography and Network Security

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1. What is Cryptography?

The science of protecting information using mathematical techniques.

- Cryptography converts readable information (plaintext) into an unreadable form (ciphertext).
- Algorithms and cryptographic keys are used to protect data from unauthorized access.
- An authorized receiver uses the correct key to recover the original message.
- Basic process: Plaintext → Encryption → Ciphertext → Decryption → Plaintext
- Applications include secure communication, online banking, passwords, digital signatures, VPNs and e-commerce.

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2. Goals of Cryptography

The fundamental security objectives of a cryptographic system.

- Confidentiality – Ensures that only authorized users can access information.
- Integrity – Ensures that data is not altered or tampered with.
- Authentication – Verifies the identity of a user, device or sender.
- Availability – Ensures reliable access to services and information when required.

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3. Common Attacks in Cryptography

Attacks can be classified into passive attacks, active attacks and other cryptographic attacks.

PASSIVE ATTACKS – The attacker observes information without modifying the system.

- Eavesdropping / Interception – Secretly listening to or capturing communication.
- Traffic Analysis – Studying communication patterns, timing and volume.

ACTIVE ATTACKS – The attacker modifies, injects or disrupts data or system operations.

- Masquerade Attack – An attacker pretends to be an authorized user.
- Replay Attack – Captured valid messages are retransmitted later.
- Message Modification – Data is altered during transmission.
- Denial of Service (DoS) – Services are disrupted or made unavailable.

OTHER ATTACKS – Brute-force, cryptanalysis, Man-in-the-Middle and side-channel attacks.

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4. Techniques to Overcome Cryptographic Attacks

Security requires strong algorithms, secure key management and correct implementation.

- Use strong, modern cryptographic algorithms and avoid outdated methods.
- Use long, random and unpredictable keys; rotate keys when required.
- Use encryption with authentication to protect confidentiality and integrity.
- Use digital certificates and authenticated communication to prevent MITM attacks.
- Use nonces, timestamps and sequence numbers to prevent replay attacks.
- Apply multi-factor authentication and strong password policies.
- Use firewalls, intrusion detection and rate limiting against DoS attacks.
- Regularly update software, patch vulnerabilities and perform security audits.

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5. From Caesar to Vigenère: Evolution of Substitution Ciphers

Substitution ciphers replace plaintext symbols with other symbols according to a rule.

Caesar Cipher – Uses one fixed shift for every letter.

Example: With shift = 3 → HELLO becomes KHOOR.

Monoalphabetic Substitution – Each plaintext letter is replaced by a fixed ciphertext letter.

Example: A → Q, B → W, C → E; the same substitution is used throughout the message.

Polyalphabetic Substitution – Uses multiple substitution alphabets to reduce frequency patterns.

Vigenère Cipher uses a repeating keyword to apply different Caesar shifts.

Vigenère Example: Plaintext = NETWORK, Key = BAD → different shifts are applied to each character.

N E T W O R K

13 4 19 22 14 17 10

B A D B A D B

1 0 3 1 0 3 1 ;

NETWORK becomes OEWXOUL

Evolution: Fixed shift → Fixed substitution alphabet → Multiple substitution alphabets.

The Vigenère cipher improves security by changing the substitution pattern throughout the message, making the frequency patterns much harder to identify than in the Caesar cipher.

Vigenère is stronger than classical Caesar and monoalphabetic ciphers

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6. From Classical to Modern Cryptography

Cryptographic techniques evolved to resist increasingly powerful attacks.

- Classical Cryptography: Caesar, substitution and transposition ciphers.
- Symmetric-Key Cryptography: The same secret key is used for encryption and decryption.
- Asymmetric-Key Cryptography: Public and private keys enable secure communication and key exchange.
- Hash Functions: Produce fixed-length digests for integrity checking and password protection.
- Digital Signatures: Provide authentication, integrity and non-repudiation.
- Modern systems combine multiple techniques to provide layered security.

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7. Conclusion

Cryptography is a continuous process of improving protection against evolving threats.

- Cryptography protects information through confidentiality, integrity, authentication and non-repudiation.
- Passive attacks observe information, while active attacks modify, inject or disrupt communication.
- Strong algorithms, secure keys, authentication and proper implementation help overcome attacks.
- Substitution ciphers evolved from the simple Caesar Cipher to the more complex Vigenère Cipher.
- Modern cryptography combines multiple techniques to provide strong, layered security.

THANK YOU

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